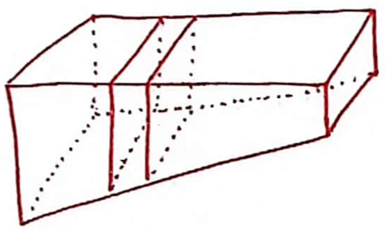
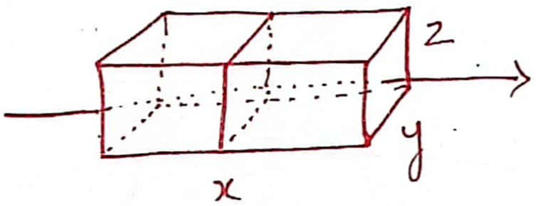
Volumes Using Cross-Sections

• Volume of Solids by Slicing:



Idea: Cut into thin slabs then use Summations to set up an integral.

To do this, find Area of Cross-Section

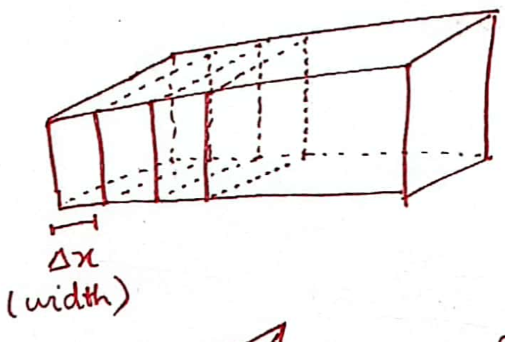


Rectangular Prism

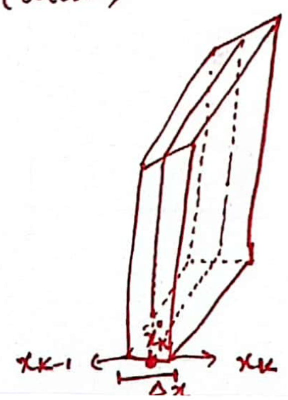
$$\text{Volume} = V = \underbrace{(y \cdot z)}_{\text{Surface area of Cross section}} \cdot \underbrace{x}_{\text{Length}}$$

* Volume of Solid along x-axis:

Now we can find the volume of any solid that is bound by planes perpendicular to x-axis at Points 'a' and 'b'.



Cut into slabs with a width of Δx



Pick arbitrary point $\overset{*}{x}_k$ on each sub-interval.

Find cross-sectional area at $\overset{*}{x}_k$.

$$\underbrace{V_K}_{\text{(volume at } K^{th} \text{ interval)}} = \underbrace{A(x_K)}_{\text{(cross-sectional area)}} \cdot \underbrace{\Delta x}_{\text{(length)}}$$

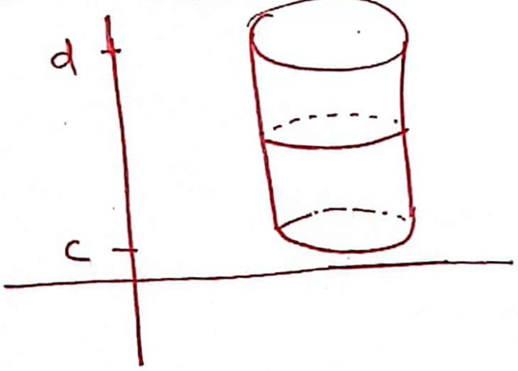
$$V = \sum_{K=1}^{n} A(x_K) \cdot \Delta x \quad \text{(Approximate volume)}$$

$$V = \lim_{n \to \infty} \sum_{K=1}^{n} A(x_K) \cdot \Delta x.$$

* $$\boxed{V = \int_a^b A(x)\, dx.}$$

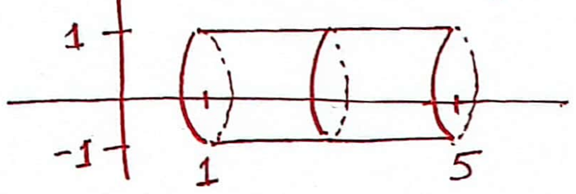
$A(x)$ = Cross-Sectional Area over $[a, b]$

* **Volume of Solid along y-axis:**



* $$\boxed{V = \int_c^d A(y)\, dy.}$$

- **Example:** Find volume of solid.



$A = \pi r^2$

$A(x) = \pi(1)^2 = \pi.$

$V = \int_1^5 \pi\, dx = \pi x \Big|_1^5$

$= \pi(5-1)$

$V = \boxed{4\pi}$

By Formula: $V = \pi r^2 h.$

$V = \pi(1)^2(5-1)$

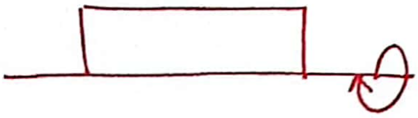
$V = \pi(1)^2(4)$

$\boxed{V = 4\pi}$

Solid of Revolution:

1

Rectangle



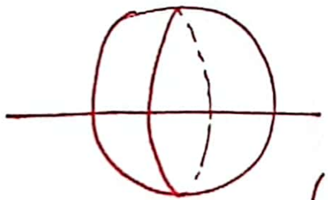
↓

cylinder (formula exists)

2

Half circle / Semi circle

↓

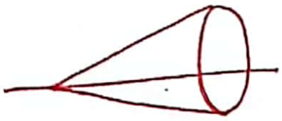


Sphere (formula exists)

3

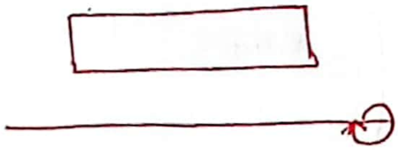
Right angled triangle

↓



cone (formula exists)

4



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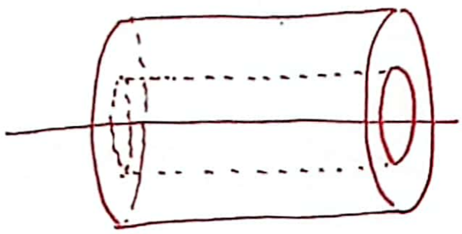
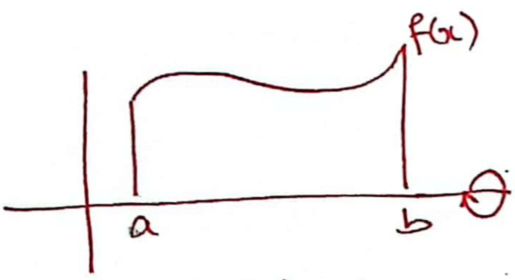
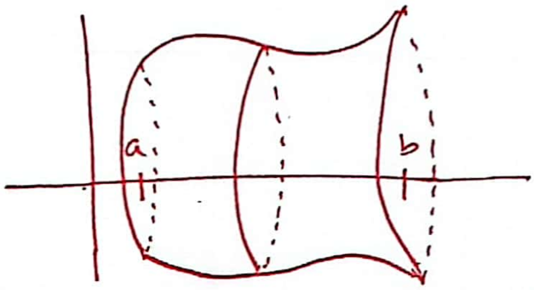


Cylinder within a cylinder. (formula exists)
(e.g tissue paper roll)

5



→

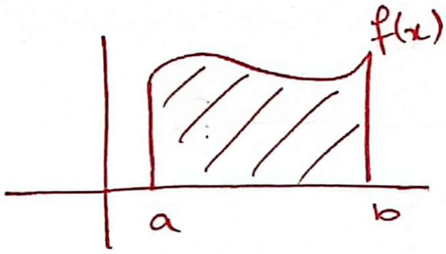


- curved edge
- irregular shape.

Q Is this a solid? Yes.

Q Are the sides of this solid perpendicular to the x-axis? Yes.

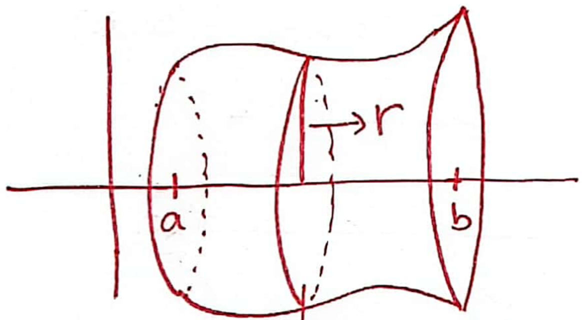
Volume of Solids of Revolution:



$f(x)$ is: (1) continuous and bounded by 'a' & 'b' ($x=a$ and $x=b$).

(2) Rotate about x-axis.

(3) Sides are $\perp$ to x-axis



cross-sectional part (circle) or disk.

Find volume by slicing:

$$V = \int_a^b A(x)\,dx$$

$A(x)$ = Surface Area of Cross-section.

$A(x) = \pi r^2$ — circle. $\quad r = f(x)$.

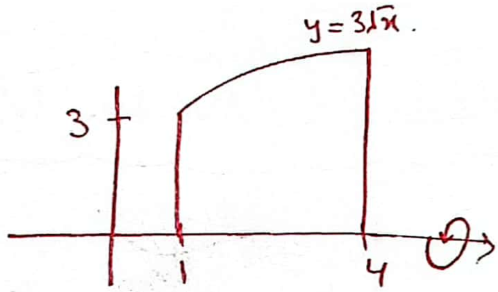
$A(x) = \pi [f(x)]^2$

* $\boxed{V = \int_a^b \pi [f(x)]^2\,dx}$ → Method of Disks

Find the volume of the solid of revolution where $y = 3\sqrt{x}$ on $[1, 4]$ is revolved about x-axis.

$\boxed{\text{or}}$

Find the volume of the solid of revolution ~~of~~ that is bound between $y = 3\sqrt{x}$, $x=1$, $x=4$ and the x-axis $(y=0)$.



Sol:

$$V = \int_a^b \pi [f(x)]^2 \, dx$$

$$= \int_1^4 \pi [3\sqrt{x}]^2 \, dx$$

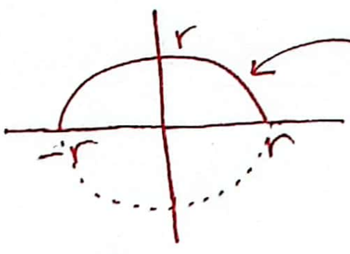
$$= \int_1^4 \pi \cdot 9x \, dx = 9\pi \int_1^4 x \, dx = 9\pi \frac{x^2}{2}\Big|_1^4$$

$$= \frac{9}{2}\pi [4^2 - 1^2] = \frac{9}{2}\pi [16-1]$$

$$= \frac{9}{2}\pi (15) = \boxed{\frac{135\pi}{2}} \text{ Ans}$$

$\longleftrightarrow$

Example 5: Derive the volume of a sphere.



$$x^2 + y^2 = r^2$$

$$y^2 = r^2 - x^2$$

$$y = \pm\sqrt{r^2 - x^2}$$

$$y = \sqrt{r^2 - x^2}$$

$$V = \int_{-r}^{r} \pi \left[\sqrt{r^2 - x^2}\right]^2 dx$$

$$= \int_{-r}^{r} \pi (r^2 - x^2) \, dx$$

$$= \pi \int_{-r}^{r} (r^2 - x^2) \, dx = \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^{r}$$

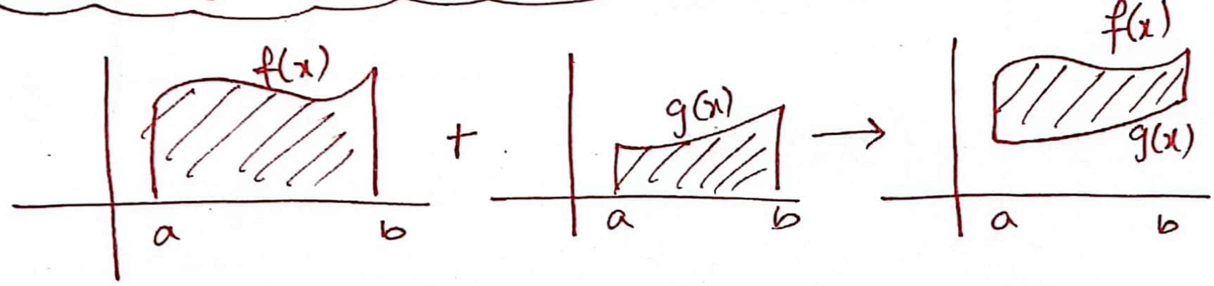
$$V = \pi\left[\left(r^2 \cdot r - \frac{r^3}{3}\right) - \left(r^2 \cdot (-r) - \frac{(-r)^3}{3}\right)\right]$$

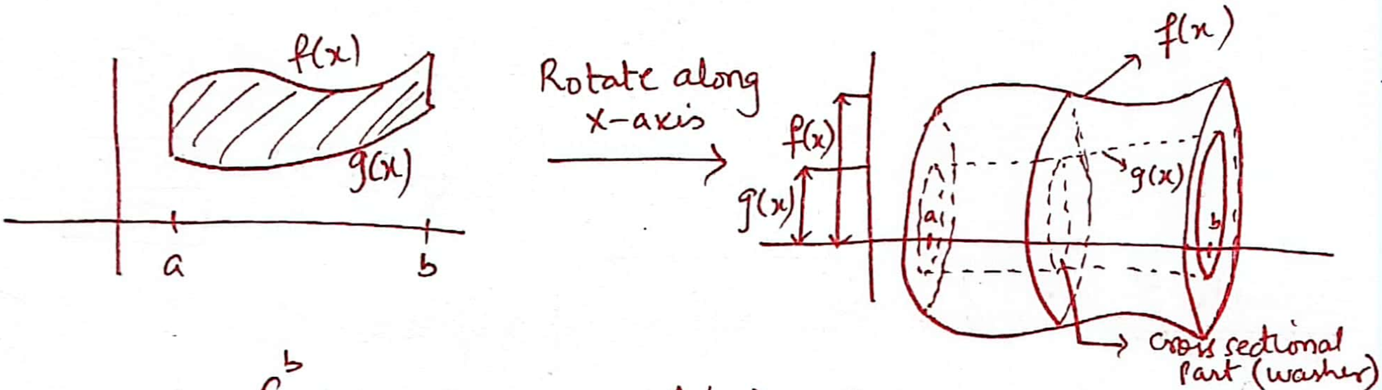
$$= \pi\left[\left(r^3 - \frac{r^3}{3}\right) - \left(-r^3 + \frac{r^3}{3}\right)\right]$$

$$= \pi\left[r^3 - \frac{r^3}{3} + r^3 - \frac{r^3}{3}\right]$$

$$= \pi\left[2r^3 - \frac{2r^3}{3}\right] = \pi\left[\frac{6r^3 - 2r^3}{3}\right] = \boxed{\frac{4}{3}\pi r^3} \text{ Ans}$$

Volume By Washer Method





$$V = \int_a^b A(x)\,dx \quad , \quad A(x) = \text{Area of Cross-Section}$$

$$A(x) = A[f(x)] - A[g(x)]$$

$$A(x) = \underbrace{\pi[f(x)]^2}_{\text{Area of large disk}} - \underbrace{\pi[g(x)]^2}_{\text{Area of Small disk.}}$$

$(\because A = \pi r^2)$

$(\because$ Radius of a circle of the cross-section is just the height of the function$)$.

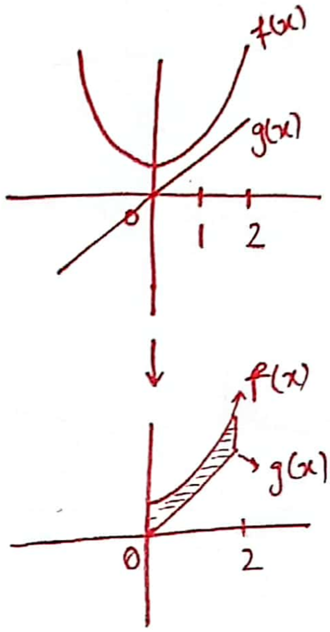
$$A(x) = \underbrace{\pi\left[[f(x)]^2 - [g(x)]^2\right]}_{\text{Area of washer}}$$

$$\star \quad \boxed{V = \int_a^b \pi\left([f(x)]^2 - [g(x)]^2\right) dx}$$

Find volume of the solid created when the area between $f(x) = \frac{1}{2} + x^2$ and $g(x) = x$ on $[0, 2]$ is rotated about x-axis.

Sol:

$$V = \int_a^b \pi\left[(f(x))^2 - (g(x))^2\right]dx.$$



$$V = \int_0^2 \pi\left[\left(\frac{1}{2} + x^2\right)^2 - (x)^2\right]dx.$$

$$V = \int_0^2 \pi\left[\frac{1}{4} + \cancel{x^2} + x^4 - \cancel{x^2}\right]dx$$

$$V = \int_0^2 \pi\left[\frac{1}{4} + x^4\right]dx$$

$$V = \pi\left[\frac{1}{4}x + \frac{x^5}{5}\right]_0^2$$

$$V = \pi\left[\left(\frac{2}{4} + \frac{2^5}{5}\right) - (0)\right] = \pi\left(\frac{1}{2} + \frac{32}{5}\right) = \pi\left(\frac{5 + 64}{10}\right)$$

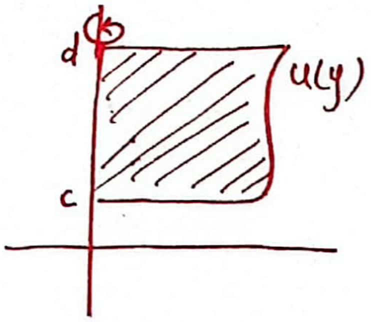
$$\boxed{V = \frac{69\pi}{10}}$$ Ans.

$\longleftrightarrow$

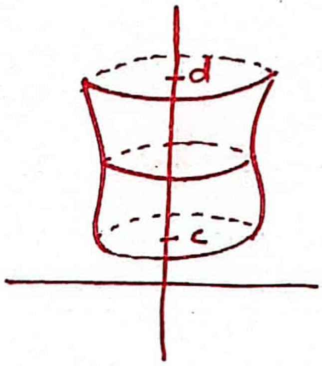
Volumes where cross-section is $\perp$ to Y-axis:

- **Disks:** $*\ \boxed{V = \int_c^d \pi[u(y)]^2\,dy}$

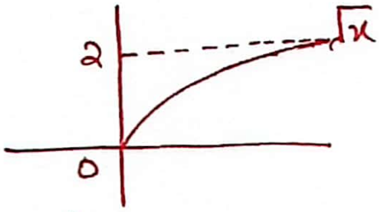
- **Washers:** $*\ \boxed{V = \int_c^d \pi\left[[u(y)]^2 - [v(y)]^2\right]dy}$



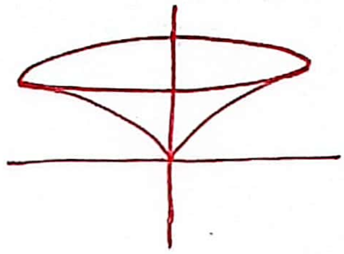
Rotate along $\longrightarrow$ y-axis



Q $y=\sqrt{x}$ is revolved around y-axis and is bound by $y=2$ and $y=0$ (x-axis). Find volume.



$\longrightarrow$

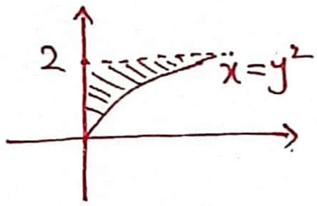


$$V=\int_c^d u(y)\,dy$$

$y=\sqrt{x}$

$x=y^2$

$$V=\int_0^2 \pi\left[y^2\right]^2 dy$$



$$V=\pi\int_0^2 y^4\,dy$$

$$V=\pi\left[\frac{y^5}{5}\right]_0^2$$

$$V=\pi\left[\frac{2^5}{5}-0\right]=\boxed{\frac{32\pi}{5}}$$ Ans

Note:

- When revolving about x-axis, put functions in terms of 'x' (Solve for 'y').
- When revolving about y-axis, put functions in terms of 'y' (Solve for 'x')

$\longleftrightarrow$

About x-axis:

$y=\sqrt{x}$

$$V=\int_0^4 \pi\left[\sqrt{x}\right]^2 dx$$

$y=\sqrt{x}$

$y=2 \rightarrow 2=\sqrt{x} \quad x=4$

$y=0 \rightarrow 0=\sqrt{x} \quad x=0$

$$=\int_0^4 \pi x\,dx=\pi\int_0^4 x\,dx=\pi\left.\frac{x^2}{2}\right|_0^4$$

$$=\pi\left[\frac{4^2}{2}-0\right]=\boxed{8\pi}$$ Ans.

$\longleftrightarrow$

Note: • The volume will be different if we rotate the curve along x-axis and y-axis.

Practice Questions

Q Find volume of the solid when the area contained by $y=x^2$ and $y=x^3$ is revolved around x-axis.

Sol:

$$V=\int_a^b \pi\left[[f(x)]^2-[g(x)]^2\right]dx$$

$$V=\int_0^1 \pi\left[(x^2)^2-(x^3)^2\right]dx$$

$$V=\int_0^1 \pi\left[x^4-x^6\right]dx$$

$$V=\pi\left[\frac{x^5}{5}-\frac{x^7}{7}\right]_0^1$$

$$V=\pi\left[\left(\frac{1^5}{5}-\frac{1^7}{7}\right)-0\right]$$

$$\boxed{V=\frac{2\pi}{35}}$$ Ans

Where do they Intercepts?

$x^3=x^2$

$x^3-x^2=0$

$x^2(x-1)=0$

$\boxed{x=0}$, $\boxed{x=1}$

Which is on top?

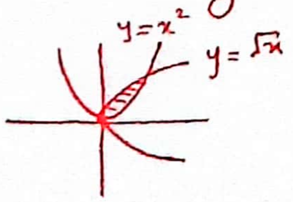
Put $x=\frac{1}{2}$

0 —— 1

Bottom $\leftarrow x^3$: $\left(\frac{1}{2}\right)^3=\frac{1}{8}$

Top $\leftarrow x^2$: $\left(\frac{1}{2}\right)^2=\frac{1}{4}$

Q Find volume of the solid when the area contained by $y=x^2$ and $x=y^2$ around y-axis.



Sol:

$$y=x^2 \rightarrow x=\sqrt{y}.$$

$$V=\int_c^d \pi\left[[u(y)]^2-[v(y)]^2\right]dy$$

Where do they intersect?

$x=\sqrt{y}$, $x=y^2$.

$\sqrt{y}=y^2$

$(\sqrt{y})^2=(y^2)^2$

$y=y^4 \Rightarrow y^4-y=0 \Rightarrow y(y^3-1)=0$, $y^3-1=0$

$\boxed{y=0}$, $\boxed{y=1}$

$$V = \int_0^1 \pi\left[(\sqrt{y})^2 - (y^2)^2\right] dy$$

$$V = \pi \int_0^1 \left[y - y^4\right] dy$$

$$V = \pi\left[\frac{y^2}{2} - \frac{y^5}{5}\right]_0^1$$

$$V = \pi\left[\left(\frac{1^2}{2} - \frac{1^5}{5}\right) - 0\right] = \pi \cdot \frac{3}{10}$$

$$\boxed{V = \frac{3\pi}{10}}$$ Ans.

When is on right?

Put $y = \frac{1}{4}$

0 1

Right ← $\sqrt{y}$: $\sqrt{\frac{1}{4}} = \frac{1}{2}$

Left ← y^2 : $\left(\frac{1}{4}\right)^2 = \frac{1}{16}$

Q Find volume of the solid when the area contained by $y = x$ and $y = x^3$ is revolved around $y = \frac{3}{2}$.

Sol:

$$V = \int_a^b \pi\left[[f(x)]^2 - [g(x)]^2\right] dx.$$

Intersect.

$x = x^3$.

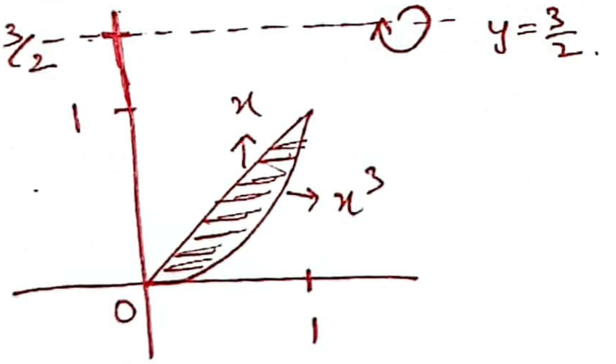
$x^3 - x = 0$

$x(x^2 - 1) = 0$

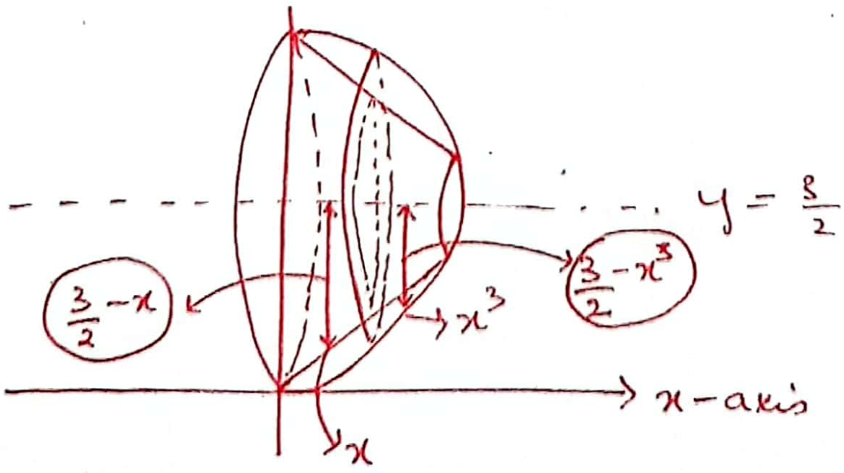
$\boxed{x = 0}$, $x = \pm 1$

$\boxed{x = 1}$

For our understanding we choose $\boxed{x = 1}$.



Rotate along ↓ x-axis.



$$V = \int_a^b \pi \left[[c - \overset{\overset{\text{outside}}{\uparrow}}{f(x)}]^2 - [c - \overset{\overset{\text{inside}}{\uparrow}}{g(x)}]^2 \right] dx$$

$$V = \int_0^1 \pi \left[\left(\frac{3}{2} - x^3\right)^2 - \left(\frac{3}{2} - x\right)^2 \right] dx$$

$$V = \pi \int_0^1 \left[\left(\frac{9}{4} - 3x^3 + x^6\right) - \left(\frac{9}{4} - 3x + x^2\right) \right] dx$$

$$V = \pi \int_0^1 \left[\cancel{\frac{9}{4}} - 3x^3 + x^6 - \cancel{\frac{9}{4}} + 3x - x^2 \right] dx.$$

$$V = \pi \int_0^1 \left[x^6 - 3x^3 - x^2 + 3x \right] dx$$

$$V = \pi \left(\frac{x^7}{7} - \frac{3x^4}{4} - \frac{x^3}{3} + \frac{3x^2}{2} \right) \Big|_0^1$$

$$V = \pi \left[\left(\frac{1^7}{7} - \frac{3(1)^4}{4} - \frac{1^3}{3} + \frac{3(1)^2}{2} \right) - 0 \right]$$

$$V = \pi \left[\frac{47}{84} \right] = \boxed{\frac{47\pi}{84}}$$ Ans.